SKILLNAVORA · PRODUCT MANAGEMENT · DESIGN FROM BASIC TO ADVANCED

Product Design for PMs: Complete Guide

How great products are designed and how PMs lead that work: research, process, UX principles, flows, visual systems, testing, accessibility, e-commerce and fintech use cases, behaviour design, metrics and interview prep.

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How to use this guide

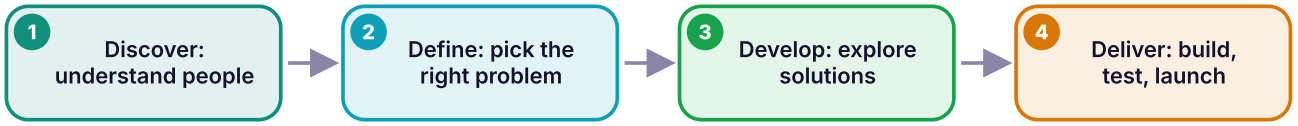
Each class has goals, concepts with Indian examples, key takeaways and a practice exercise, from basic to advanced. Finish with the final project to build something you can show in interviews.

	Class	Level
Class 1	**What product design is, and what a PM owns**	Basic
Class 2	**Know your users: research methods and when to use them**	Basic
Class 3	**The product design process, step by step**	Basic
Class 4	**Core UX principles every PM must know**	Basic
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Class 6	**Interaction design: flows, states, forms and microcopy**	Intermediate
Class 7	**Visual design and design systems for PMs**	Intermediate
Class 8	**Prototyping and usability testing**	Intermediate
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Class 11	**Fintech product design: use cases and how it works**	Intermediate
Class 12	**Other industries: delivery, mobility, SaaS, health and education**	Intermediate
Class 13	**Behaviour design, growth and the ethics of persuasion**	Advanced
Class 14	**Measuring design: metrics, experiments and impact**	Advanced
Class 15	**Scaling design: systems, collaboration and design debt**	Advanced
Class 16	**AI-era product design**	Advanced
Class 17	**Product design interviews: how to answer and what interviewers want**	Advanced
Class 18	**Final project: a product design case study**	Advanced
	Toolkit: templates, product design cheat sheet, glossary, 30 interview questions, further reading	

BASIC · CLASS 1

What product design is, and what a PM owns

Understand the craft, the team and the PM's role.



The Double Diamond: diverge, converge, twice

Product design is deciding *how a product works and feels* so that real people can solve a real problem easily. It is more than visuals. It covers research, structure, flows, interface, words, behaviour, testing and the way all of this is built and measured.

Layer (Garrett's five planes)	Question	E-commerce example
Strategy	Why does this exist? Who is it for?	Help busy shoppers find and buy the right product fast
Scope	What features and content?	Search, filters, reviews, wishlist, returns
Structure	How is it organised and how do users move?	Categories, product page, cart, checkout flow
Skeleton	Where does each element go?	Wireframes: where the price, the delivery date and the buy button sit
Surface	How does it look and sound?	Colours, type, photos, button labels

Who does what

PM	Designer	Engineer
Problem, goals, priorities, trade-offs, success metrics	Research, flows, interface, prototypes, usability	Feasibility, performance, building, quality

> The PM does **not** pick fonts or draw screens. The PM makes sure the team solves the right problem, for the right user, with a clear measure of success, and keeps design, engineering and business aligned.

Key takeaways

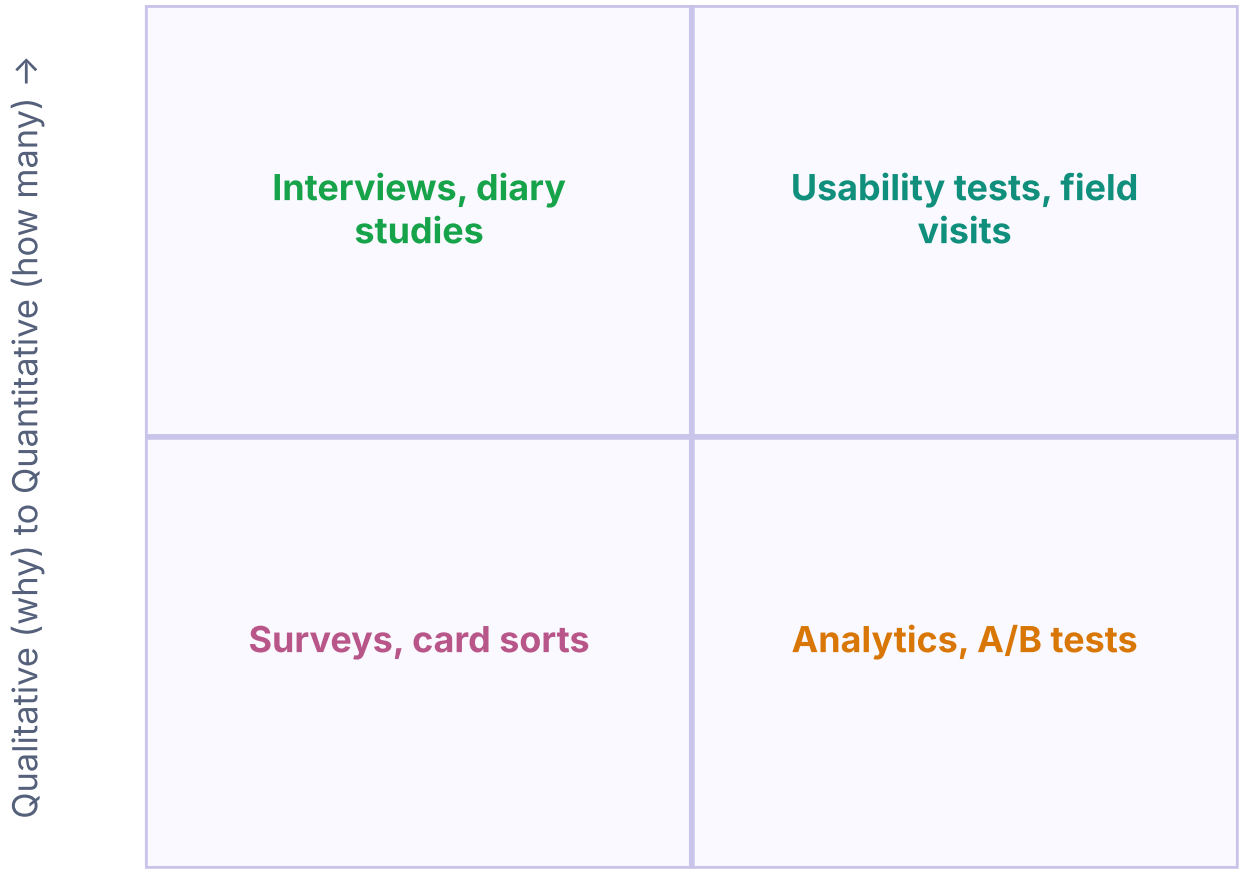
- Product design covers the whole experience, not only looks.
- Five planes: strategy, scope, structure, skeleton, surface.
- PM owns the problem and outcomes; designer owns the solution craft; both own the user.

Practice exercise: Pick an app you use daily and write its strategy, scope and structure in one line each.

BASIC · CLASS 2

Know your users: research methods and when to use them

Choose the right way to learn from users.



Research methods map

Method	Use it to	Example
User interviews	Understand needs, motives and context	Ask 8 shoppers how they choose a phone, and what made them abandon the last cart
Usability testing	See where people get stuck	Watch 5 first-time investors try to complete KYC
Surveys	Measure opinions at scale	CSAT after a delivery
Analytics	See what people do, and where they drop	Funnel from product page to payment
Session recordings, heatmaps	Spot confusion and rage taps	Users tapping a non-clickable image
Card sort / tree test	Fix navigation and labels	Where would you look for "returns"?
Support tickets and reviews	Find recurring pain	Top 10 complaint themes

Turning research into design inputs

- **Jobs To Be Done (JTBD):** "When I [situation], I want to [motivation], so I can [outcome]." Example: "When I run out of milk at 8 am, I want it in 10 minutes, so I can make breakfast."
- **Persona:** a short profile of a user type based on real data, not a made-up character.
- **Journey map:** the steps, feelings and pain points across time.
- **Insight format:** observation, reason, opportunity. Not "users want a faster checkout" but "6 of 8 users abandoned at address entry because typing long addresses on a phone is slow; offer map pin and autofill".

> Five users find most big usability problems in one round (Nielsen Norman Group). Test early with few people, fix, then test again.

Key takeaways

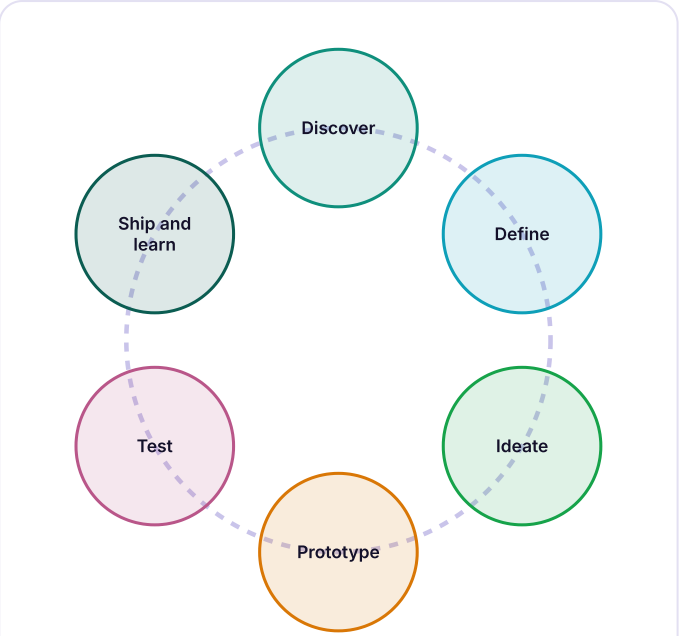
- Mix qualitative (why) and quantitative (how many).
- JTBD and journey maps turn research into design direction.
- Test small and often.

Practice exercise: Plan a 5-person usability test for the checkout of an app you know: goals, tasks and what you will measure.

BASIC · CLASS 3

The product design process, step by step

Follow a repeatable path from problem to launch.



The design loop

Stage	What happens	Output the PM reviews
1. Discover	Research users, data, competitors, constraints	Insights and opportunity areas
2. Define	Frame the problem and the success measure	Problem statement, "How might we...?", metric target
3. Ideate	Explore many solutions before choosing	Sketches, flows, 3 options with trade-offs
4. Prototype	Make something testable	Clickable Figma prototype
5. Test	Usability tests, feedback, small experiments	Findings and changes
6. Build and ship	Specs, design QA, release, monitor	Shipped feature and dashboards
7. Learn	Measure outcomes, iterate	Impact review, next bets

Useful formats

- **Problem statement:** "[User] struggles to [goal] because [barrier], which causes [business impact]."
- **How Might We:** turns a pain into an open question: "How might we make address entry take under 10 seconds?"
- **Design sprint (5 days):** map, sketch, decide, prototype, test, for a big question in one week.
- **Lean version:** a 1 to 2 week loop: one assumption, one prototype, five tests.

The most common failure is jumping from problem to a favourite solution. Insist on at least three options and a clear reason for the pick.

Key takeaways

- Design is a loop, not a handoff.
- Define the problem and metric before screens.
- Compare at least three options.

Practice exercise: Write a problem statement and three 'How might we' questions for reducing cart abandonment.

BASIC · CLASS 4

Core UX principles every PM must know

Learn the rules of thumb that predict good and bad design.

Nielsen's 10 usability heuristics (with examples)

Heuristic	Good example	Bad example
Visibility of system status	"Order packed, rider on the way", with live map	Spinner with no message after paying
Match with the real world	"Add to bag", "Wallet"	"Instantiate transaction"
User control and freedom	Undo send, cancel order	No way to remove a coupon
Consistency and standards	Same button position and wording on every screen	"Submit", "Done", "OK" for the same action
Error prevention	Confirm before sending a large UPI payment; disable Pay while processing	Easy double-tap on Pay
Recognition over recall	Saved addresses, recent searches	Retyping card details every time
Flexibility and efficiency	Shortcuts, reorder in one tap	Same long flow for every visit
Aesthetic, minimalist design	One main action per screen	Five competing banners
Help users recover from errors	"Card declined: your bank rejected it. Try UPI?"	"Error 4021"
Help and documentation	Contextual tips, searchable FAQ	A 40-page PDF

Laws of UX in plain words

Law	Meaning	Design use
Hick's law	More choices, slower decisions	Limit options; group and filter
Fitts's law	Bigger, closer targets are easier to hit	Large thumb-reach buttons (at least 44 to 48 px)
Miller's law	People hold about 7 items in short-term memory	Chunk numbers: 4-4-4-4 card input
Jakob's law	Users expect your app to work like others they know	Use familiar patterns (cart icon top right)
Von Restorff effect	The different item is remembered	One highlighted primary button
Peak-end rule	People judge by the peak and the ending	Delight on success screens; smooth refunds
Doherty threshold	Responses under about 400 ms feel instant	Skeleton screens, optimistic updates
Zeigarnik effect	Unfinished tasks stay on the mind	Progress bars, "complete your profile 80%"

Key takeaways

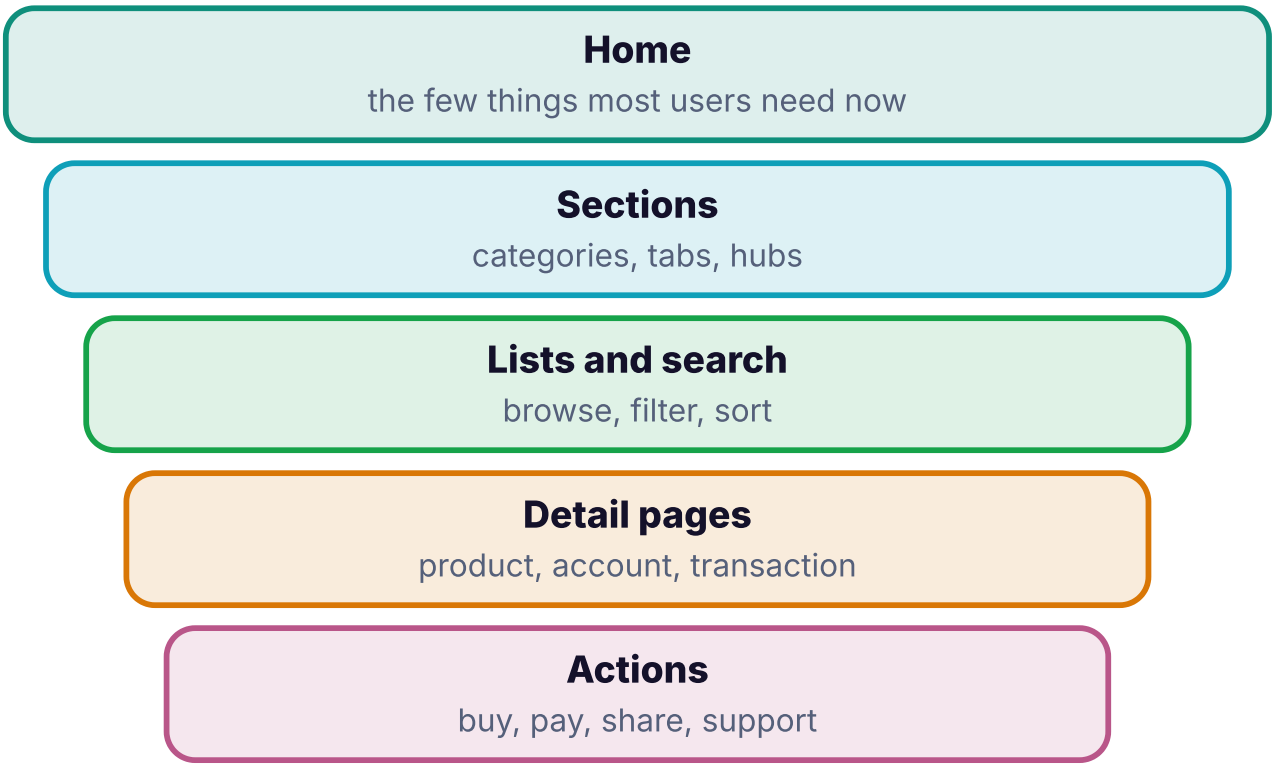
- Heuristics are a fast review checklist.
- Fewer choices and bigger targets improve completion.
- Feedback, forgiveness and familiarity beat cleverness.

Practice exercise: Review one checkout against the 10 heuristics and list the 3 worst violations.

INTERMEDIATE · CLASS 5

Information architecture and navigation

Organise content so people can find things without thinking.



Information architecture from broad to specific

Information architecture (IA) is how content and features are grouped, labelled and connected. Good IA means users find what they need in few steps, using words they would use.

Pattern	When it works	Example
Bottom tab bar (3 to 5 items)	Frequent switching between main areas on mobile	Home, Orders, Wallet, Profile
Search-first	Huge catalogue	Marketplaces, grocery
Hub and spoke	Distinct tasks from one place	Banking super-app: pay, recharge, invest
Hamburger or drawer	Rarely used items only	Settings, legal pages
Wizard / stepper	Long, linear tasks	KYC, loan application

How to design and test IA

- **Card sorting:** ask users to group cards (products or features) and name the groups.
- **Tree testing:** give users a text-only menu and a task; measure if they reach the right place.
- **Labels:** use customer words ("Refund status") not internal ones ("Reverse logistics").
- **Depth:** aim for the most common task in 3 taps or fewer.

Example: An e-commerce app that hid "Track order" under Profile, then Orders, then Details saw many "where is my order?" support calls. Moving a live order card to the Home screen reduced the calls and added a trust signal.

Key takeaways

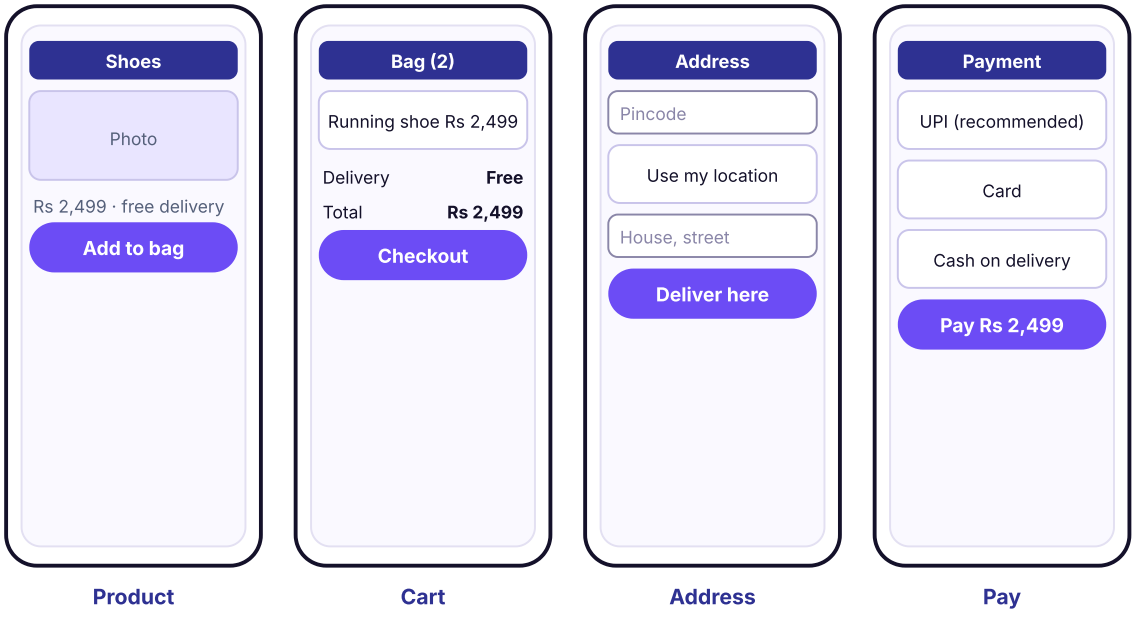
- IA is grouping, labelling and linking.
- Three to five bottom tabs, search for big catalogues.
- Test with card sorts and tree tests, not opinions.

Practice exercise: Redesign the top-level navigation for a food delivery app and test it with three friends.

INTERMEDIATE · CLASS 6

Interaction design: flows, states, forms and microcopy

Design the moments between the screens.



A four-screen purchase flow: one primary action per screen

- **User flow:** the path from a goal to its completion, with decisions and exits. Draw the happy path first, then errors and edge cases.
- **Screen states:** every screen needs **empty** ("No orders yet, browse deals"), **loading** (skeleton), **error** ("No internet. Try again"), **partial**, **success**, and **offline**.
- **Forms:** ask only what you need, one column, correct keyboard type, inline validation, smart defaults, autofill, show password option.
- **Microcopy:** short, specific, human. "Pay Rs 2,499" beats "Submit". Say what happens next.
- **Feedback and motion:** acknowledge every tap within 100 ms; use motion to explain, not decorate.
- **Progressive disclosure:** show the essentials, reveal detail on demand.
- **Onboarding:** let users reach value fast; ask permissions in context ("Allow location to show nearby stores").

Bad copy	Better copy
Invalid input	Enter a 10-digit mobile number
Transaction failed	Payment not completed. Your money is safe. Try again or use UPI.
Are you sure? Yes / No	Delete this address? Delete / Keep

> Ask your designer for the "state sheet": every state of every screen. Missing states cause most production bugs and support tickets.

Key takeaways

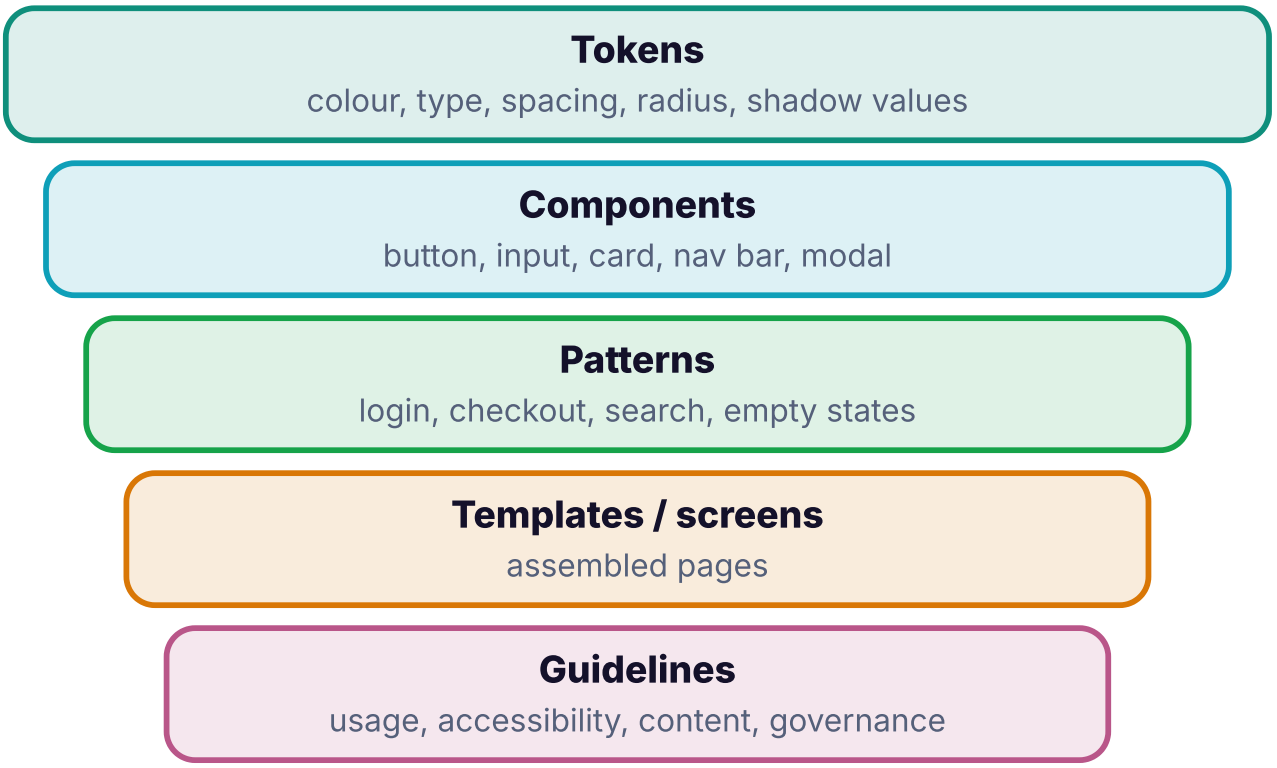
- Design flows and all states, not only the happy path.
- One primary action per screen; specific microcopy.
- Reduce form effort with defaults, autofill and the right keyboard.

Practice exercise: List all states for an order tracking screen and write the microcopy for each.

INTERMEDIATE · CLASS 7

Visual design and design systems for PMs

Read a screen like a designer and know what a design system buys you.



Layers of a design system

Concept	What to look for
Hierarchy	The eye should land on the most important thing first (size, weight, colour, space)
Grid and spacing	Consistent 4 or 8-pixel steps; alignment; breathing room
Typography	Two typefaces at most, 3 to 4 sizes, readable body text (at least 16 px on mobile)
Colour	One brand colour for actions; meaning colours (red errors, green success); never colour alone
Contrast	Text contrast at least 4.5:1 (WCAG AA)
Imagery and icons	Real, consistent, with labels where meaning is unclear
Dark mode and themes	Separate palette, not an inverted copy

Why design systems matter

- **Speed:** teams assemble screens from ready parts.
- **Consistency:** the whole product feels like one product.
- **Quality and accessibility** are built into components once.
- **Cheaper change:** change a token (the brand blue) once, update everywhere.

Examples to study: Google Material Design, Apple Human Interface Guidelines, Atlassian, Polaris (Shopify), and India examples like Razorpay Blade and Zomato Sushi.

> PM tip: when a designer says "this is not in the system", ask the cost of a new component versus reusing an existing one. Custom UI is slower to build, test and maintain.

Key takeaways

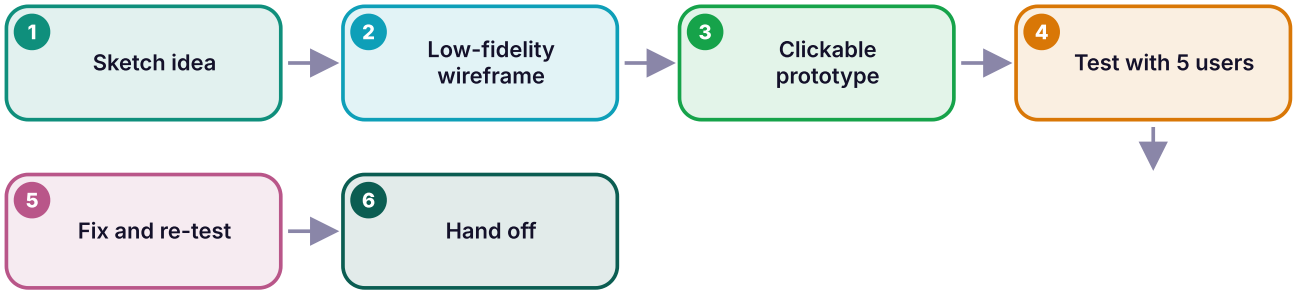
- Hierarchy, spacing, contrast and consistency make most of the difference.
- A design system is tokens, components and patterns with rules.
- Reuse beats reinvent unless the user value is clear.

Practice exercise: Audit three screens for hierarchy and contrast and write five fixes.

INTERMEDIATE · CLASS 8

Prototyping and usability testing

Test ideas cheaply before engineers build them.



From idea to tested design

Fidelity	Use	Tool
Paper or whiteboard	Compare ideas in minutes	Pen, Miro
Low-fi wireframe	Test structure and flow, no visual distraction	Figma, Balsamiq
High-fi prototype	Test look, copy and interactions	Figma, ProtoPie
Coded prototype / beta	Test real data and performance	Lovable, Replit, feature flags

A simple usability test

1. **Goal:** "Can new users buy a phone in under 3 minutes?"
2. **Tasks:** "Find a phone under Rs 20,000 and buy it with UPI." Do not give hints.
3. **Think aloud:** ask users to say what they are thinking.
4. **Observe:** task success, time, errors, confusion, quotes.
5. **Prioritise:** severity = how many users and how bad (blocker, major, minor).
6. **Report:** top 5 issues, evidence clips, proposed fixes.

- **Measures:** task success rate, time on task, error rate, the **SUS** questionnaire (System Usability Scale, 68 is about average), and satisfaction.
- **Usability test vs A/B test:** usability tests explain *why* with few people; A/B tests measure *how much* with many. Use both.
- **Remote tools:** Maze, UserTesting, Lookback, or a simple video call with screen share.

> Do not ask "Do you like it?". Ask people to **do** something and watch. Opinions are weak evidence; behaviour is strong.

Key takeaways

- Match fidelity to the question.
- Five users, real tasks, think aloud.
- Usability tests find why; A/B tests find how much.

Practice exercise: Write a test script with three tasks for a prototype you have, and run it with two people.

INTERMEDIATE · CLASS 9

Accessibility and inclusive design (with India in mind)

Design for everyone, including low-end phones and many languages.

Area	What to do	Why
Contrast	Text at least 4.5:1; large text 3:1	Low vision, sunlight
Touch targets	At least 44 by 44 pt (iOS) or 48 by 48 dp (Android)	Motor impairments, big thumbs
Screen readers	Labels for every control, logical order, alt text	Blind and low-vision users (TalkBack, VoiceOver)
Text scaling	Layout works at 200% font size	Older users
Not colour alone	Icons and text with status colours	Colour blindness
Captions and transcripts	For audio and video	Deaf users, noisy places
Reduced motion	Respect system setting	Vestibular conditions

Designing for India and emerging markets

- **Languages:** Hindi and regional languages, mixed-language text, longer strings that break layouts, voice input.
- **Low-end phones and poor networks:** small app size, lite mode, skeleton screens, offline states, compressed images.
- **Trust and literacy:** icons plus text, clear amounts in rupees, no jargon, help in the user's language, human support.
- **Shared devices:** families sharing one phone; protect payments with PIN or biometrics.
- **Payments:** UPI-first, COD option for e-commerce, clear pending and failed states.

> Accessibility is also a business and legal issue: more customers, better SEO and usability for everyone, and compliance in many markets (WCAG 2.2 AA is the common target; India's Rights of Persons with Disabilities Act applies to digital services).

Key takeaways

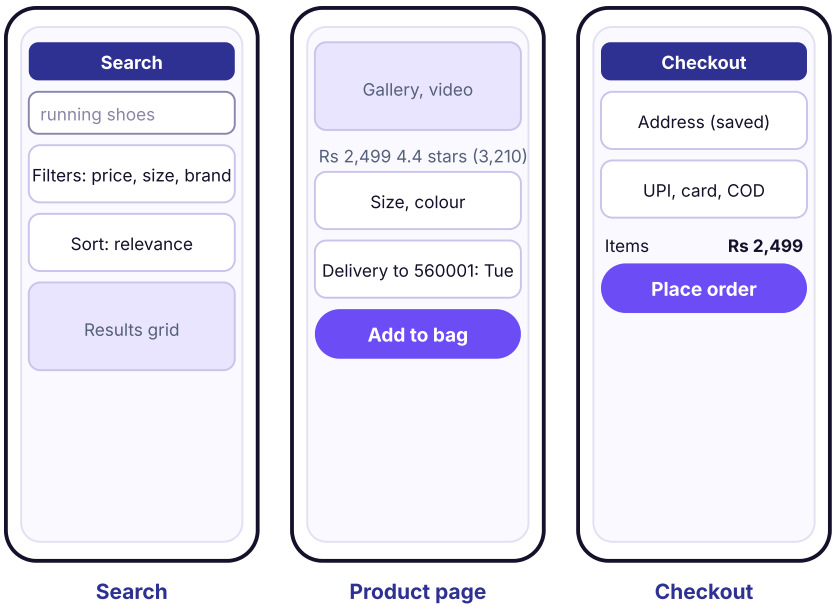
- Accessibility benefits all users and the business.
- Contrast, targets, labels, text scaling are the basics.
- In India, design for language, low-end devices and low-trust contexts.

Practice exercise: Run an accessibility check on your favourite app: contrast, target size, font scaling and screen reader labels.

INTERMEDIATE · CLASS 10

E-commerce product design: use cases and how it works

Design discovery, product pages, cart and checkout that convert and build trust.



E-commerce core screens

Stage	Use case	How the design works	Metric
Discovery	"I need running shoes under Rs 3,000"	Search with autosuggest and typo tolerance, filters that match how people decide (size, price, rating), good sort defaults, category pages, personalised rows	Search success, zero-result rate, click-through
Product page	"Is this the right one, and can I trust it?"	Big photos and video, price and offers visible, size guide, delivery date by pincode, reviews with photos, return policy, "add to bag" always visible	Add-to-cart rate, bounce
Cart	"What will I pay in total?"	Show full cost early (delivery, taxes), edit quantity, save for later, coupon field that does not distract, stock warnings	Cart-to-checkout rate
Checkout	"Pay quickly and safely"	Guest or one-tap login by OTP, saved address, address autofill, UPI first, trust badges, one primary button, clear errors	Checkout completion, payment success rate
After purchase	"Where is my order? I want to return it."	Live tracking, proactive updates, easy returns and refunds	Support contacts per 100 orders, repeat rate

What research says

Across 50 studies, Baymard Institute's list puts the average documented cart abandonment rate at about **70%**, and its checkout research finds the average large store has dozens of fixable usability

problems. The usual causes: surprise costs, forced account creation, long or confusing forms, limited payment methods and trust worries. Source: Baymard Institute.

Quick commerce and marketplace twists

- **Quick commerce:** speed promise (10 minutes), live stock, minimal choice per category, one-screen cart, accurate ETA, substitutions policy.
- **Two-sided marketplaces:** design for buyer and seller; seller tools (listing, pricing, inventory, payouts) need the same care as the buyer app.
- **Trust:** genuine reviews, clear seller info, easy return policy, secure payment signs.

> **Common mistakes:** forced sign-up before checkout, delivery cost revealed at the last step, hidden coupon fields that send users off to search for codes, tiny tap targets, and the "Pay" button that can be tapped twice.

Key takeaways

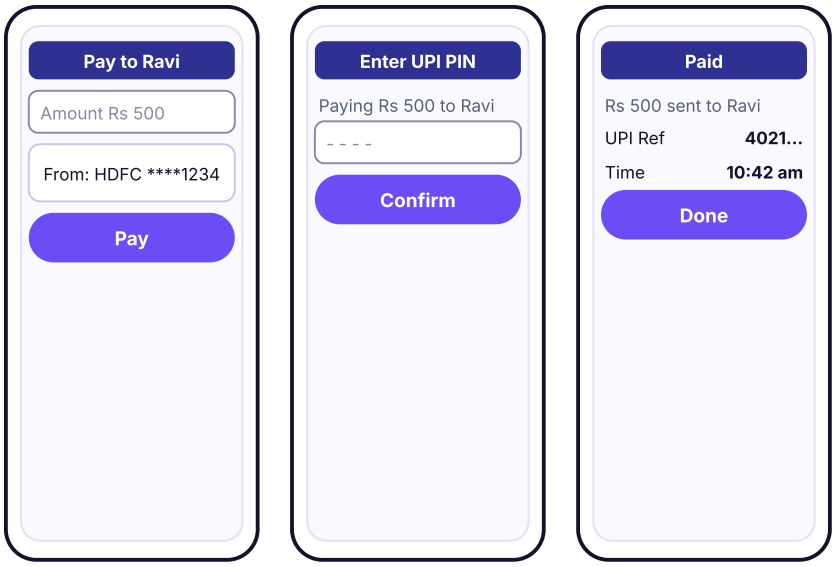
- Every stage has a user question to answer.
- Show total cost and delivery date early; fewer fields at checkout.
- Measure each stage: search, add-to-cart, checkout, payment success, returns.

Practice exercise: Write the top five checkout friction points of any shopping app and propose one fix with a metric for each.

INTERMEDIATE · CLASS 11

Fintech product design: use cases and how it works

Design money moments that feel safe, clear and fast.



A UPI payment: three steps, each with clear status

Use case	How it works	Design principles
Onboarding and KYC	Mobile OTP, PAN, Aadhaar eKYC or video KYC, bank link	Explain why each item is needed, show progress, save and resume, allow camera retry tips, fast approval status
Pay with UPI	Pick payee, enter amount, authenticate with UPI PIN, bank confirms, receipt	Verify the payee name before paying, protect against wrong amounts, show "pending" honestly, never hide the reference number
Cards and wallets	Tokenised card, 3-D Secure OTP, wallet balance	Never ask for card details twice, show masked numbers, clear failure reasons and "money is safe" copy
Lending and BNPL	Eligibility check, offer, agreement, disbursal, EMI	Show full cost (rate, fees, total repayable) before consent; no dark patterns; reminders before due dates
Investing	Risk profile, product choice, order, portfolio	Plain-language risk warnings, no gamified over-trading, long-term framing, regulated disclosures (SEBI)
Insurance	Quote, buy, claim	Simple coverage summary, easy claim steps and status

Fintech design rules

- **Clarity of money:** amounts, fees and totals in plain numbers, with currency, never hidden.

- **Trust and security:** biometrics, PIN, device binding, visible security cues, fraud warnings ("Do not share OTP").
- **Honest status:** success, pending, failed, with what to do next and when money returns. Failed-but-debited is the most stressful state; design it with care.
- **Error prevention:** confirm payee name, limit warnings, recent-payee safety checks.
- **Compliance by design:** consent screens, data localisation, RBI and SEBI rules, DPDP Act privacy. Collaborate with legal early.
- **Fraud-aware UX:** friction in the right place (new payee, large amount), not everywhere.

Metrics: KYC completion and time, first-transaction rate, payment success rate by method and bank, failure reasons, support contacts, fraud rate, activation (first money in) and retention.

Key takeaways

- Money UX is about clarity, trust and honest status.
- Design the pending and failed states most carefully.
- Add friction only where risk is high.

Practice exercise: Draw the 'failed but money debited' screen for a UPI payment and write the copy.

INTERMEDIATE · CLASS 12

Other industries: delivery, mobility, SaaS, health and education

See the same principles applied to different services.

Industry	Core use case	How the design works	Key design focus
Food delivery	Order dinner and know when it arrives	Search by cuisine and restaurant, menu with photos, customise items, live tracking of order and rider, ETA updates	Accurate ETA, trust, reorder in one tap, clear fees
Ride hailing	Get a cab in minutes	Pickup pin, fare estimate, driver match, live map, OTP to start, rating	Reassurance while waiting, safety (SOS, share trip), fare transparency
B2B SaaS	Complete work faster	Dashboard, tables, filters, bulk actions, roles and permissions, onboarding checklists, empty states with sample data	Efficiency, learnability, time to value, enterprise needs
Healthcare	Book a doctor and manage records	Doctor search and slots, video visit, prescriptions, reminders, reports	Privacy, clarity, accessibility, reduce anxiety
Education	Learn and stay consistent	Course path, lessons, quizzes, progress, streaks, community	Motivation, progress visibility, short sessions
Travel and tickets	Book seats or rooms	Search by date and place, filters, price transparency, seat map, hold timer, payment, e-ticket	Price clarity, hold and timeout design, urgency without false claims
Social and content	Discover and share content	Feed, follow, create, notifications, safety tools	Relevance, creators' needs, notification control, well-being

Patterns that repeat across industries

- **Search, filter, sort:** food, products, doctors, flights.
- **Live status and ETA:** delivery, rides, orders, payments.
- **Trust cues:** ratings, verified badges, secure labels, clear policies.
- **Reorder and shortcuts** for repeat users.
- **Empty, loading and error states** everywhere.

> When designing for a new industry, learn the **vocabulary**, the **risks** (money, safety, health) and the **regulation** first. Then reuse proven patterns instead of inventing new ones.

Key takeaways

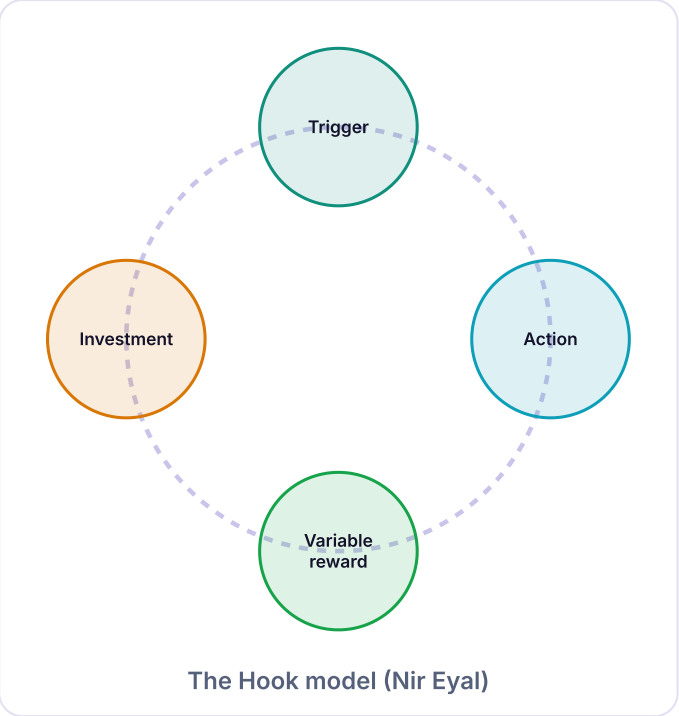
- The same principles apply across industries.
- Learn the domain's risks and rules before designing.
- Reuse proven patterns: search, status, trust, repeat.

Practice exercise: Pick an industry you do not know. List its three biggest user risks and one design pattern for each.

ADVANCED · CLASS 13

Behaviour design, growth and the ethics of persuasion

Use habit and motivation design without manipulating users.



The Hook model (Nir Eyal)

Concept	Meaning	Example
Fogg behaviour model	Behaviour = motivation x ability x prompt	Make the action easy (one tap), prompt at the right time
Hook model	Trigger, action, variable reward, investment	Notification, open app, new content, save preferences
Streaks and goals	Loss aversion, progress	Duolingo streak; fitness streak with a weekly freeze
Social proof	People follow others	"1,200 bought this week"
Defaults and nudges	The default is chosen most often	Pre-selected UPI, a recommended plan
Activation	The first moment of real value	First order delivered, first investment made

Dark patterns to avoid (they hurt trust and invite regulation)

- **Fake urgency** ("only 2 left" when untrue) and **drip pricing** (fees added at the end).
- **Confirmshaming** ("No, I do not want to save money").
- **Forced continuity** and hard-to-cancel subscriptions; **basket sneaking**; **disguised ads**.
- India's Central Consumer Protection Authority has issued guidelines on dark patterns, and the DPDP Act requires clear consent.

Test: would you be comfortable showing this design and its metric to your user's family, a regulator and the press? If not, redesign it.

Key takeaways

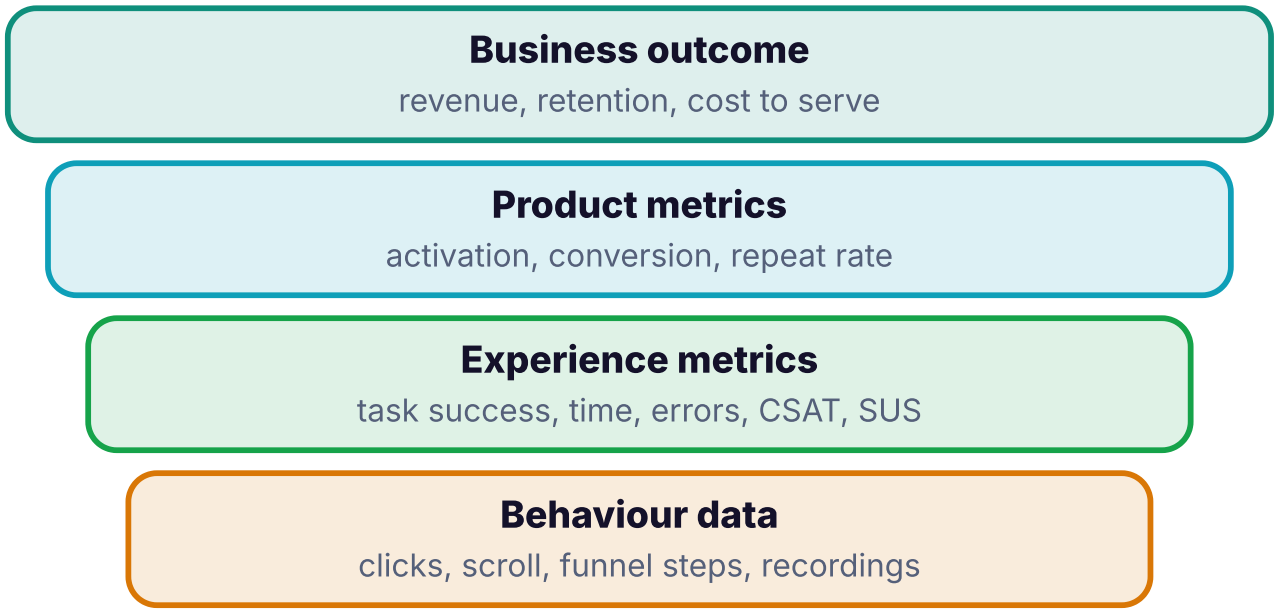
- Make the right behaviour easy and well-prompted.
- Habits come from value plus a repeatable loop.
- Avoid dark patterns; trust is a long-term metric.

Practice exercise: Redesign a subscription cancel flow that is clear and fair, and note what metric you would watch.

ADVANCED · CLASS 14

Measuring design: metrics, experiments and impact

Prove that design work moves outcomes.



Metrics ladder from behaviour to business

Framework	What it contains
HEART (Google)	Happiness, Engagement, Adoption, Retention, Task success; define goals, signals and metrics for each
Funnel analysis	Steps from entry to goal; find the biggest drop and ask why
Usability benchmarks	SUS, task success, time on task, error rate, tracked release over release
A/B testing	Compare versions with random users; one clear primary metric, guardrails, enough sample
Qualitative signals	Support tickets, reviews, interviews, session recordings

Example: redesigning checkout

- **Hypothesis:** showing delivery date and total cost earlier will increase checkout completion.
- **Primary metric:** checkout completion rate. **Guardrails:** payment success rate, refund rate, page speed.
- **Design:** run an A/B test on 50% of traffic for two full weeks; segment by new vs returning users and by device.
- **Read-out:** lift with confidence interval, plus qualitative review of drop-offs.
- **Decision:** ship, iterate or roll back.

> Not every design change can be A/B tested (small traffic, big navigation shifts). Then use usability studies, staged rollouts, before-and-after comparisons and customer feedback, and be honest

about uncertainty.

Key takeaways

- Link design changes to outcome metrics and guardrails.
- Use HEART to cover happiness and task success, not only clicks.
- Combine experiments with qualitative understanding.

Practice exercise: Define goals, signals and metrics (HEART) for a new onboarding flow.

ADVANCED · CLASS 15

Scaling design: systems, collaboration and design debt

Run design well across teams and over time.

- **Working with designers:** bring problems, constraints and data; not solutions. Share the goal and success metric early. Review in critiques; give feedback as user impact, not taste ("users may miss this action because it blends with the background").
- **Design critique:** present the goal, show options, ask specific questions, record decisions.
- **Handoff and design QA:** annotated specs, states, tokens, responsive behaviour; check the built UI against design before release.
- **Design debt:** inconsistent components, old patterns and missing states. Track it like tech debt; reserve capacity; link fixes to metrics (speed, error rate, conversion).
- **Design ops:** tools, libraries, research repository, rituals, consistent processes.
- **Scaling across platforms:** iOS, Android, web, tablet, foldables; shared logic, platform-specific behaviour.
- **Localisation:** languages, formats, right-to-left text, cultural symbols.
- **Build vs buy vs adapt** for components and design kits.

Situation	Good PM response
Designer and engineer disagree on effort	Ask for options and trade-offs, pick by user impact and cost, document the decision
Stakeholder wants a banner on the home screen	Ask the goal; test impact on core tasks; propose placement and rules
Deadline pressure	Cut scope, not quality of core flow; use a smaller first release
Design feels inconsistent	Audit components, fix the top 5, add to the system

Key takeaways

- Bring problems and data to design, not solutions.
- Give feedback as user impact.
- Manage design debt like tech debt.

Practice exercise: Write a design critique brief for a feature you are working on: goal, options, questions.

ADVANCED · CLASS 16

AI-era product design

Design AI features that users can trust and control.

- **Set expectations:** say what the AI can and cannot do; show confidence or sources where useful.
- **Keep the human in control:** preview, edit, undo, approve before actions with consequences.
- **Handle errors gracefully:** AI will be wrong sometimes; give easy ways to correct it and report issues.
- **Conversational UI:** suggestions and quick replies reduce blank-page anxiety; combine chat with structured UI (cards, forms) for actions like payments.
- **Personalisation with privacy:** explain why a recommendation is shown; offer controls and opt-outs.
- **Latency and streaming:** show progress; stream answers.
- **Safety:** guardrails, escalation to humans for sensitive topics (money, health).

Industry	AI design example
E-commerce	Visual search, size advice, review summaries, an assistant that compares products with sources
Fintech	Spend insights, fraud alerts with explanation, a support assistant that hands over to humans for disputes
Support	RAG chatbot with citations and a "talk to an agent" button

See the Designing AI Experiences (AI UX) course for the full method.

Key takeaways

- Set expectations, show sources, keep humans in control.
- Design for AI mistakes, not only success.
- Combine chat with structured UI.

Practice exercise: Design the error and correction flow for an AI shopping assistant.

ADVANCED · CLASS 17

Product design interviews: how to answer and what interviewers want

Practise the common formats with a clear structure.

Question type	Example	What they test
Design X for Y	"Design a savings app for college students."	User empathy, structure, prioritisation, trade-offs
Improve an existing product	"How would you improve Swiggy's search?"	Critique, problem finding, metrics
Critique	"What do you like and dislike about this checkout?"	Design vocabulary, user thinking
Open-ended object	"Design an alarm clock for the blind."	Constraints, creativity, inclusion
Collaboration	"A designer disagrees with your priority. What do you do?"	Communication, influence, outcomes
Metrics	"How would you measure a redesign's success?"	Goal-metric-guardrail thinking

A structure you can use (state it out loud)

1. **Clarify:** product, platform, scope, goal, constraints. Ask 3 or 4 questions.
2. **Users:** pick 2 or 3 segments, choose one, and say why.
3. **Needs and pain points:** top 3, in their words.
4. **Prioritise:** pick one problem by impact and effort.
5. **Solutions:** 3 options, a pick with reasons.
6. **Flow and key screens:** describe the happy path and one edge case, with states.
7. **Metrics:** goal, success metric, guardrails.
8. **Risks and next steps:** experiments, rollout, what you would learn first.

Worked answer: "Design a savings app for college students"

- **Clarify:** India, 18 to 24, Android first; goal: first-time savers.
- **Users:** hostel students with irregular pocket money; part-time earners. Pick the first.
- **Pains:** money disappears on small spends, no habit, distrust of apps, no minimum balance.
- **Prioritise:** a painless way to save small amounts automatically.
- **Options:** round-up savings on UPI spends; a savings goal with friends; a weekly fixed auto-save.
- **Pick:** round-ups plus a visible goal ring; trust through bank partners and clear withdrawals anytime.
- **Flow:** link UPI, set goal, see round-ups, celebrate milestones; states for failed link and low balance.
- **Metrics:** activation (first round-up), weekly active savers, average saved per month; guardrails: withdrawals within 24 hours, complaints, overdraft.

Tips: think aloud, use real product examples, talk about trade-offs, name risks, keep to the structure but be flexible when the interviewer redirects, and end with metrics and learning plan.

Key takeaways

- Use a clear structure and say it out loud.
- Choose one user and one problem, justified.
- Always finish with metrics, risks and next steps.

Practice exercise: Record a 5-minute answer to 'Design a savings app for college students' using the structure.

ADVANCED · CLASS 18

Final project: a product design case study

Create a portfolio-quality case that shows your thinking.

Pick one real problem in e-commerce, fintech or another service you use (examples: reduce checkout drop-off, make UPI failures less stressful, improve order tracking). Produce a 10-slide or 2-page case with:

1. Problem, user and goal with the evidence (analytics, reviews, 5 interviews).
2. Current journey map and the three biggest friction points.
3. Three solution options and the pick.
4. Key screens (sketch or Figma), all states and microcopy.
5. Accessibility and inclusion check.
6. Test plan: prototype test with 5 users and a proposed A/B test.
7. Metrics: primary, secondary, guardrails, expected impact.
8. Risks, trade-offs and rollout plan.

Present it for five minutes to a friend, record their questions and update the case.

Key takeaways

- A good case shows problem, evidence, options and impact.
- Show trade-offs and what you would test.
- Practise presenting the story in five minutes.

Practice exercise: Complete the case study and share it with a designer for feedback.

TOOLKIT

Templates

Problem statement

Field	Fill in
User and context	
Goal they want to achieve	
Barrier or pain (with evidence)	
Business impact	
Success metric	
Constraints	

Usability test plan

Field	Fill in
Goal	
Participants (5) and recruitment	
Tasks (3)	
Success criteria	
What we will measure	
Findings and decisions	

Design review checklist

Field	Fill in
Goal and metric clear?	
Happy path and edge cases?	
All states (empty, loading, error, offline)?	
Accessibility (contrast, targets, labels)?	
Microcopy reviewed?	
Analytics events defined?	
Trust and compliance reviewed?	

Product design interview answer

Field	Fill in
Clarify	
Users and segment	
Pain points	
Priority problem	
Three options and the pick	
Flow and states	
Metrics and guardrails	
Risks and next steps	

TOOLKIT

Product design cheat sheet

Term / tool	Meaning / use
Double Diamond	Discover, Define, Develop, Deliver
JTBD	Jobs To Be Done: situation, motivation, outcome
IA	Information architecture: grouping and labelling content
Heuristic evaluation	Expert review against usability principles
Hick's law	More choices, slower decisions
Fitts's law	Large, near targets are easier to hit
Jakob's law	Users expect familiar patterns
Peak-end rule	People remember the peak and the end
Wireframe	Low-fidelity layout of a screen
Prototype	Clickable model for testing
Design system	Tokens, components, patterns and rules
Design token	A named design value such as a colour or spacing step
SUS	System Usability Scale questionnaire
HEART	Happiness, Engagement, Adoption, Retention, Task success
A/B test	Random comparison of two versions
Dark pattern	A deceptive design that tricks users
WCAG	Web Content Accessibility Guidelines; AA is the common target
Skeleton screen	Placeholder shape shown while content loads
Progressive disclosure	Reveal detail only when needed
Empty state	Screen with no data yet, with guidance

Glossary

Affordance	A cue showing how something can be used
Cognitive load	Mental effort needed to use an interface
Information scent	Cues that tell users a link leads to what they want
Mental model	How users think a system works
Persona	Research-based profile of a user type
Journey map	Steps, feelings and pain points over time
User flow	Path from goal to completion
Card sorting	Users group items to reveal their mental model
Tree testing	Testing a text-only menu for findability
Think-aloud	Users speak thoughts while using a product
Task success rate	Share of users who complete a task
Hamburger menu	Hidden navigation icon with three lines
Bottom navigation	Tab bar at the bottom of a mobile screen
Microcopy	Small interface text such as buttons and errors
Responsive design	Layout adapts to screen sizes
Fidelity	How close a mockup is to the final product
Onboarding	Guiding new users to first value
Activation	The moment a new user first gets real value
Friction	Anything that slows or stops a user
Conversion rate	Share of users who complete a goal
Drop-off	Users leaving at a funnel step
Nudge	A gentle push toward a choice
Confirmshaming	Guilt-based opt-out wording
Accessibility	Usable by people with a wide range of abilities
Touch target	Tappable area of a control
Contrast ratio	Difference between text and background colour
Component	Reusable interface part
Design debt	Accumulated inconsistencies and shortcuts
Handoff	Passing design details to engineering
Design QA	Checking the built UI against the design

INTERVIEW PREP

30 interview questions with answers

Q1. What is product design, and how is it different from UI design?
Product design covers the whole experience: problem, research, structure, flows, interface, content, testing and measurement. UI design is the visual layer inside it.

Q2. What is the role of a PM in product design?
Define the problem, users, goals and metrics, set constraints and priorities, enable research and testing, and align design, engineering and business. The PM does not dictate the visuals.

Q3. Walk me through your design process.
Discover users and data, define the problem and metric, ideate several options, prototype, test with users, ship, measure and iterate.

Q4. How do you decide which problem to solve?
Combine evidence from users and data, size the impact on the business goal, check feasibility and strategic fit, and prioritise by impact versus effort.

Q5. Design a savings app for college students.
Clarify scope, pick a user (hostel students), name top pains, prioritise automatic small savings, compare options (round-ups, goals, auto-save), pick one, describe flow and states, and define activation and retention metrics with guardrails.

Q6. How would you improve checkout on an e-commerce app?
Look at the funnel, find the biggest drop, show total cost and delivery date earlier, reduce form fields with saved addresses and autofill, offer UPI first, add trust cues, prevent double payment, and measure completion and payment success.

Q7. What makes a good fintech UX?
Clarity of money, honest status, security cues, error prevention, plain language, and compliance by design. Handle pending and failed states with care.

Q8. A payment failed but money was debited. What should the screen say?
State what happened, that the money is safe, when it returns or will be confirmed, a reference number, and the next action. Keep the order pending until the gateway confirms.

Q9. How do you reduce cart abandonment?
Remove surprises (costs, forced sign-up), simplify forms, offer more payment options, add trust, recover with reminders, fix speed, and use research to find the real reasons.

Q10. How do you measure a redesign's success?
Set a primary outcome metric, supporting experience metrics (task success, time), and guardrails. Use A/B tests where possible, plus usability studies and qualitative signals.

Q11. Usability test or A/B test?
Usability tests with few users explain why people struggle; A/B tests measure how much a change matters at scale. Use both.

Q12. How many users do you need for a usability test?
About five users find most major issues in a round; repeat with new users after fixes.

Q13. Explain Hick's law with an example.
More options increase decision time. A checkout with ten payment methods shown equally is slower

than showing the top three and grouping the rest.

Q14. What are dark patterns? Give examples.
Deceptive designs such as fake urgency, hidden fees, confirmshaming and hard-to-cancel subscriptions. They harm trust and attract regulation.

Q15. How do you design for accessibility?
Meet contrast and target size, label controls for screen readers, support text scaling, do not rely on colour alone, caption media, and test with assistive tech.

Q16. How do you design for low-end phones and poor networks in India?
Small app size, lite modes, skeleton screens, offline states, compressed images, vernacular languages, voice input and clear status messages.

Q17. What is a design system and why does it matter?
Reusable tokens, components and patterns with rules. It speeds delivery, improves consistency and accessibility, and lowers cost of change.

Q18. A designer disagrees with your priorities. What do you do?
Listen, restate the goal and evidence, ask for their user-impact reasoning, explore options and trade-offs, decide with data, and document it.

Q19. A stakeholder wants a promotional banner on the home screen.
Clarify the business goal, test the impact on core task completion, propose limits (placement, frequency), and measure with guardrails.

Q20. How do you prioritise design debt against new features?
Quantify the user and business impact (conversion, errors, speed, support tickets), reserve a regular share of capacity, and bundle fixes with related feature work.

Q21. What is the Double Diamond?
A model with two diverge-converge phases: discover and define the problem, then develop and deliver the solution.

Q22. What are JTBD and how do you use them?
A way to describe what users want to achieve in a situation. Use it to frame problems and compare alternatives, including non-product alternatives.

Q23. How do you design an onboarding flow?
Get users to first value fast, ask only essential info, ask permissions in context, show progress, let users skip, and measure activation.

Q24. How would you improve search in a food delivery app?
Study queries and zero-result searches, add autosuggest and typo tolerance, support cuisine, dish and restaurant intents, show relevant filters, personalise, and measure search success and conversion.

Q25. How do you design for trust in a marketplace?
Verified sellers, real reviews with photos, clear policies, secure payments, transparent pricing and quick support.

Q26. How should an AI assistant be designed in a shopping app?
Set expectations, show sources, let users edit and undo, combine chat with product cards, give easy escalation and handle errors gracefully.

Q27. What is the HEART framework?
Google's framework: Happiness, Engagement, Adoption, Retention, Task success, each with goals, signals and metrics.

Q28. How would you test a prototype before engineering builds it?
Create a clickable prototype, give 5 users realistic tasks, observe, fix, and repeat; then plan a staged rollout or A/B test.

Q29. Describe a design you would remove from a product.
Choose a feature that adds clutter or friction without value (for example a forced sign-up wall), explain the evidence and the metric you would watch after removing it.

Q30. What would you do in your first 30 days as PM on a design-heavy product?
Meet users and designers, review research and analytics, audit key flows against heuristics, find top friction, agree on goals and metrics, and ship one small improvement.

LEARN MORE

Further reading and tools

- Nielsen Norman Group: 10 usability heuristics: https://www.nngroup.com/articles/ten-usability-heuristics/
- Laws of UX: https://lawsofux.com/
- Baymard Institute: cart abandonment research: https://baymard.com/research/cart-abandonment-rate
- Baymard: checkout usability benchmark: https://baymard.com/research/checkout-usability
- Design Council: the Double Diamond: https://www.designcouncil.org.uk/our-resources/the-double-diamond/
- Google Material Design 3: https://m3.material.io/
- Apple Human Interface Guidelines: https://developer.apple.com/design/human-interface-guidelines/
- W3C WCAG overview: https://www.w3.org/WAI/standards-guidelines/wcag/
- The Design Sprint (book site): https://www.thesprintbook.com/
- NN/g: why 5 users are enough: https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/
- Figma Learn: https://help.figma.com/hc/en-us
- Refactoring UI (visual design for developers and PMs): https://www.refactoringui.com/
- Google HEART framework (research paper): https://research.google/pubs/measuring-the-user-experience-on-a-large-scale-user-centered-metrics-for-web-applications/
- Deceptive design (formerly dark patterns): https://www.deceptive.design/

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